

Coupled iterative loop — repeat until $\Delta A < 1\%$, $\Delta E_a < 1\%$

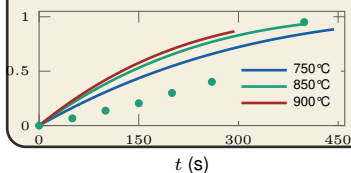
(a) TGA data & problem

$$\frac{dX}{dt} = A \exp\left(\frac{-E_a}{RT}\right) f(X)$$

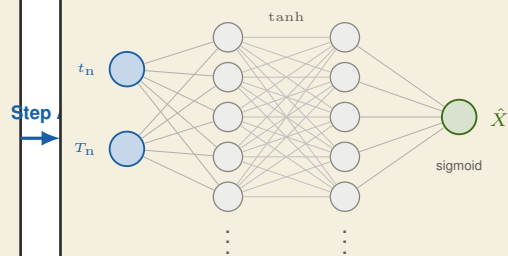
$$X(0) = 0$$

$$T \in \{750, 850, 900\}^\circ\text{C}$$

8 obs. per temperature

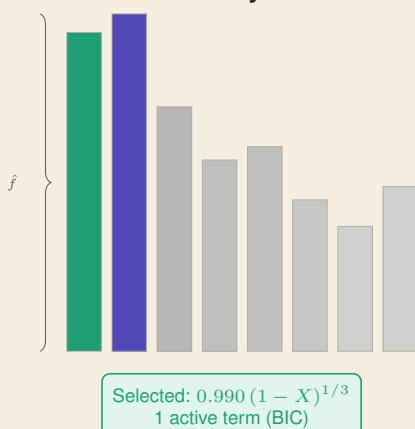


(b) Inverse PINN



Trainable: $\log A$, $\log E_a$
jointly optimised with weights

(c) Library



(d)

$$\mathcal{L} = \underbrace{\lambda_{\text{phys}} \frac{1}{N_c} \sum_{i=1}^{N_c} \left(\left. \frac{dX}{dt} \right|_{\text{PINN}} - A^{(k)} \exp\left(\frac{-E_a^{(k)}}{RT_i}\right) f^{(k)}(X_i) \right)^2}_{\text{physics residual (Step A)}} + \underbrace{\lambda_{\text{data}} \frac{1}{N_{\text{obs}}} \sum_{j=1}^{N_{\text{obs}}} (X_j^{\text{pred}} - X_j^{\text{obs}})^2}_{\text{data loss (24 sparse obs.)}} + \underbrace{\lambda_{\text{ic}} \frac{1}{N_T} \sum_T \hat{X}(0, T)^2}_{\text{initial condition}}$$

(e)

Converged results — iteration 8 ($\Delta A = 0.33\%$, $\Delta E_a = 0.08\%$)

Forward validation

Arrhenius parameters

	A	E_a
Wang 2023 (JMA)	—	10.3–26.7 kJ/mol
Preprint 1 (SCM)	$5.87 \times 10^{-2} \text{ s}^{-1}$	24.07 kJ/mol
This work	$7.50 \times 10^{-2} \text{ s}^{-1}$	27.75 kJ/mol

Discovered $f(X)$

$$0.990 (1 - X)^{1/3}$$

grain model — 1 active term

No prior model assumed

